ESTR1002 Fall 2025

Lab Exercise #3

Due: 1 Oct 2025

Separate your answer for each question in a different .c file. Compress all files into a .zip and submit it to Blackboard.

Question 1

Write a program to print a giraffe whose neck length is defined by an input integer number. You are not required to use loops.

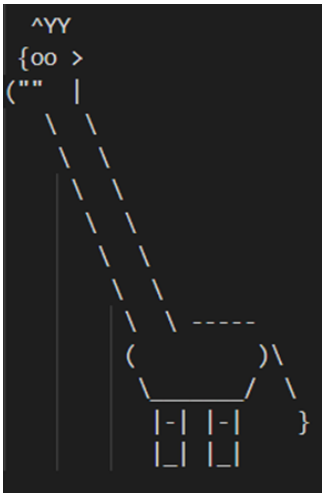
Input: an integer in the range [3, 7].

Output: some text print-outs as specified below.

Sample Run #1:
Input: 3
Output:



Sample Run #2:
Input: 7
Output:



Question 2

Write a program that requests the number of hours one worked in a month and then prints out the gross pay, taxes, and net pay, where gross pay = taxes + net pay.

Assume that the gross pay are calculated as follow:

- Basic pay rate = $75.50 per hour
- Overtime (>160 hours) = one and a half time of the basic pay rate

And the taxes are:

- Tax rate = 10% for first $10000 of gross pay, 16% for next $3000, and 23% for the rest.

Input: an integer x in [0, 500] that represents the number of hours one worked in a month.

Output: print out the gross pay, taxes, and net pay, rounded to 2 decimal places.

Note: for accuracy, please use “double” for all variables.

The sample output is shown as follows (the underlined characters are the inputs from the user).

Sample Run #1:

```
0
gross  = 0.00
tax    = 0.00
netpay = 0.00
```

Sample Run #2:

```
100
gross  = 7550.00
tax    = 755.00
netpay = 6795.00
```

Sample Run #3:

```
321
gross  = 30313.25
tax    = 5462.05
netpay = 24851.20
```

Question 3

Given the start and arrival time of a train, write a program to calculate the total train traveling time (which is assumed to be no more than 24 hours).

Input: two four-digit integers, separated by a space. The first integer is the departure time in the format of “HHMM”, where HH ranges [0, 23] and MM ranges [0, 59] denote hour and minute. The second integer is the arrival time, which is also represented in the same “HHMM” format.

Output: an integer ranges [1, 24x60], representing the train traveling time in minutes.

The sample output is shown as follows (the underlined characters are the inputs from the user).

Sample Run #1:

```
1001 1320
199
```

Sample Run #2:

```
0650 1303
373
```

Sample Run #3:

```
2345 0055
70
```

Question 4

RGB (Red, Green, Blue) and HSV (Hue, Saturation, Value) are two color formats for encoding colors in computers. Each color is written as triplets in both RGB and HSV. For example, the pure red is written as (255, 0, 0) in RGB, or equivalently, (R=255, G=0, B=0). In HSV, the pure red is (0.0, 1.0, 1.0), or in the form of (H=0.0, S=1.0, V=1.0).

Now, given a color written in RGB, please write a C program to convert it into its HSV equivalent. The program should take three integers ranging from 0 to 255 as the user-input RGB values, perform the conversion according to the process below, and print out the HSV values as three floating numbers.

Here is the conversion process:

Step 1: Normalize the RGB values: Convert the RGB components from a range of 0-255 to 0-1 by dividing each by 255.

Step 2: Determine the maximum and minimum of the RGB values: This helps in calculating the Value (V) and the range of color intensity.

Step 3: Calculate Value (V): V is the maximum of the normalized R, G, and B values.

Step 4: Calculate Saturation (S):

- If V is 0, S is set to 0, as it implies a lack of color intensity (dark/black).
- Otherwise, S is calculated by $(V - \min)/V$, where $\min$ is the minimum of the normalized RGB values.

Step 5: Calculate Hue (H):

- If S is 0, H is set to 0, which typically corresponds to a grayscale tone.
- For colors (when S is not 0), H depends on which RGB component is the maximum:
  - **If Red is max**, $H = (G - B)/(V - \min)$
  - **If Green is max**, $H = 2.0 + (B - R)/(V - \min)$
  - **If Blue is max**, $\mathrm{H} = 4.0 + (\mathrm{R} - \mathrm{G})/(\mathrm{V} - \min)$
  - Note that the G, B, R here are all normalized (between 0 and 1). If multiple RGB components share the same value as V, you can consider any of them as the maximum value.
- Multiply H by 60 to convert it to degrees. If H is negative, add 360 to ensure it falls within the 0 to 360 range. Convert H from a range of 0-360 to 0-1 by dividing it by 360.

Input: three integers, representing the RGB values of a color. Each value is between 0 and 255.

Output: Print the corresponding HSV values (each rounded to 3 decimal):

Hue (H) as a float between 0 and 1.

Saturation (S) as a float between 0 and 1.

Value (V) also as a float between 0 and 1.

The sample output is shown as follows (the underlined characters are the inputs from the user).

Sample Run #1:

```
205 192 176
0.092
0.141
0.804
```

Sample Run #2:

```
240 248 255
0.578
0.059
1.000
```

Question 5

A group of scientists have a close encounter with a group of aliens. Since the scientists believe that number is the universal language across the universe, they try to talk with the aliens through numbers.

When the scientists display **the decimal number 31** to the alien, the aliens seem to understand and one of them gives a visual response: **a number 25**. The scientists immediately understand that each of the aliens may be using a different number system than us (and assume that the scientist is using the decimal number system).

The scientists guess that the alien is translating 31 in decimal into their number system, and then the scientist knows that the **alien is using a number system of base 13**. However, the alien may be cheating: they give a random number, e.g., 7, which is not possible to represent 31 in decimal under any number systems.

Write a program that helps the scientists (1) either find the base of the number system the alien is using, or (2) determine that the alien is cheating.

Input: two integers in the range of [1, 999], inclusively. The first integer is from the scientists and the second integer is from the aliens.

Output: if the alien is not cheating, print the **lowest-possible base** of the number given by the alien. Else, print the phrase "**CHEATED**".

The sample output is shown as follows (the underlined characters are the inputs from the user).

Sample Run #1:

```
31 25
13
```

Sample Run #2:

```
25 31
8
```

Sample Run #3:

```
31 1
CHEATED
```

Hints: How does the number system work in maths? Formulate the maths problem!